

Positive slopes

Lines that slope upward from left to right have positive slopes. The equation of a line with a positive slope can be worked out from its graph, as described below.

Find the slope of the line by choosing a section of it and drawing horizontal (green) and vertical (red) lines out from it so they meet. Count the units each new line covers, then divide the vertical by the horizontal distance.

$$\text{slope} = \frac{\text{vertical distance}}{\text{horizontal distance}} = \frac{6}{3} = +2$$

+ sign means line slopes upward from left to right

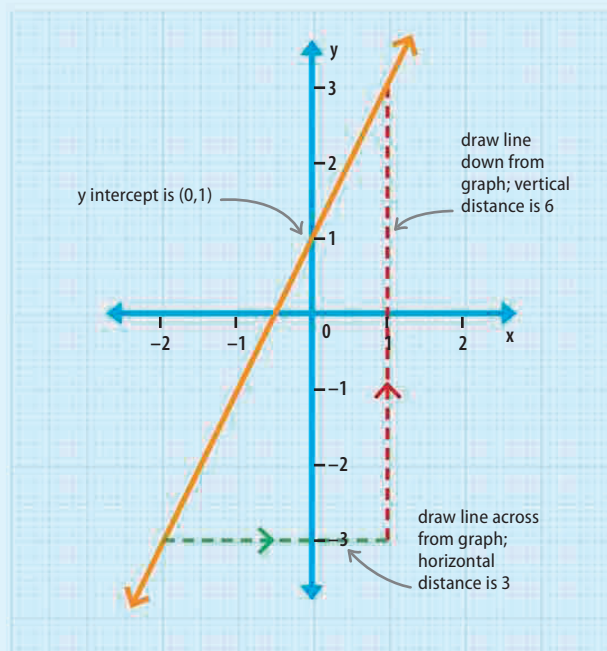
The y intercept can be easily read off the graph—it is the point where the line crosses the y axis.

$$\text{y intercept} = (0,1)$$

Substitute the values for the slope and y intercept into the equation of a line to find the equation for this given line.

y intercept is 1 slope is +2

$$y = mx + b \Rightarrow y = 2x + 1$$



Negative slopes

Lines that slope downward from left to right have negative slopes. The equation of these lines can be worked out in the same way as for a line with a positive slope.

Find the slope of the line by choosing a section of it and drawing horizontal (green) and vertical (red) lines out from it so they meet. Count the units each new line covers, then divide the vertical by the horizontal distance.

$$\text{slope} = \frac{\text{vertical distance}}{\text{horizontal distance}} = \frac{4}{1} = -4$$

insert minus sign to show line slopes downward from left to right

The y intercept can be easily read off the graph—it is the point where the line crosses the y axis.

$$\text{y intercept} = (0,-4)$$

Substitute the values for the slope and y intercept into the equation of a line to find the equation for this given line.

y intercept is (0,-4) slope is -4

$$y = mx + b \Rightarrow y = -4x - 4$$

